Thad Avery

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Education

University of Texas at Austin Austin, TX
Master of Science in Artificial Intelligence and Machine Learning *Aug. 2024 – Aug. 2025*

- Relevant Coursework: Advanced Machine Learning, Deep Learning, Reinforcement Learning, Natural Language Processing, AI Ethics, Model Interpretability, Adversarial Robustness

Colorado State University Fort Collins, CO
Bachelor of Science in Computer Science, Minor in Chemistry *Aug. 2019 – May 2023*

Experience

Software Engineer Feb. 2024 – Present
Tru Solutions *Remote*

- Integrated AI-powered tools like Devin, Cursor, and Cypress with AI to streamline workflows and automate test coverage, directly improving development efficiency
- Applied advanced knowledge of AI from my academic studies to identify and implement intelligent solutions, aligning with modern software practices
- Achieved an 80% increase in automation testing coverage, significantly enhancing the app's reliability and reducing manual effort
- Proficient in React, JavaScript, and MySQL, with a proven ability to quickly learn and apply new skills
- Contributed to the success of a small team, demonstrating adaptability and a strong work ethic

Grand Canyon River Guide June 2015 – Aug. 2022
Hatch River Expeditions *Grand Canyon National Park, AZ*

- Led and facilitated week-long excursions for groups of 30 people, navigating the Colorado River in Grand Canyon National Park
- Managed complex logistics and safety protocols while operating 36-foot motorboats in challenging conditions
- Prioritized and enforced comprehensive backcountry safety through clear communication and orientations
- Delivered engaging interpretive presentations on geology, ecology, and cultural history
- Demonstrated exceptional problem-solving and decision-making skills in high-stakes environments

Projects

Transformer-Based Language Text Model | *PyTorch, Python* 2024

- Designed and implemented a Transformer encoder from scratch, achieving over 95% accuracy on a character-level sequence prediction task
- Developed a language model achieving perplexity of 6.3 within 10 minutes of training on Wikipedia text data
- Explored model interpretability by analyzing attention mechanisms, aligning with AI safety research areas
- Implemented advanced techniques such as positional encodings, causal masking, and multi-head attention mechanisms

Deep Averaging Networks for Sentiment Classification | *PyTorch, NLP* 2024

- Built a deep averaging network using PyTorch to classify sentiment in text data, achieving over 77% accuracy
- Integrated pre-trained GloVe embeddings and implemented fine-tuning for improved performance
- Enhanced model robustness against noisy data using prefix embeddings and spelling correction techniques, relevant to adversarial robustness research

Technical Skills

Machine Learning: Neural Networks, Transformers, Model Interpretability, Adversarial Robustness, PyTorch, TensorFlow, Scikit-learn, NLP
Extras: Completed Andrej Karpathy's AI tutorials on neural networks and deep learning
Languages: Python, C, C++, Java, JavaScript, TypeScript, SQL
Frameworks & Tools: React, Redux, Node.js, Git, Agile Development, Unit Testing
Web Development: HTML, CSS, Frontend, Backend, Full-Stack